



# ME 200

# Thermodynamics

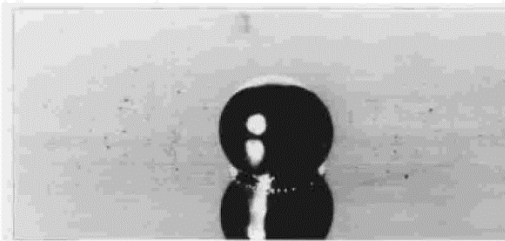
Jiahuan CUI, PhD

Lecture 03 – Work

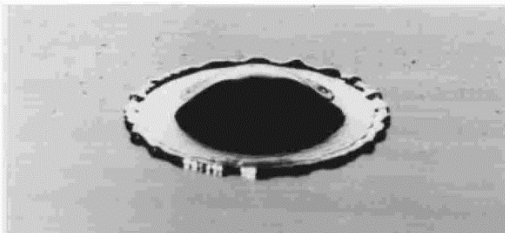
# Surface Tension Work

240°C, 2 m/s

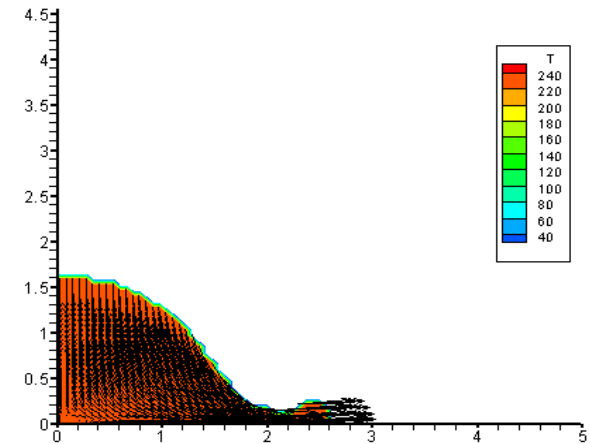
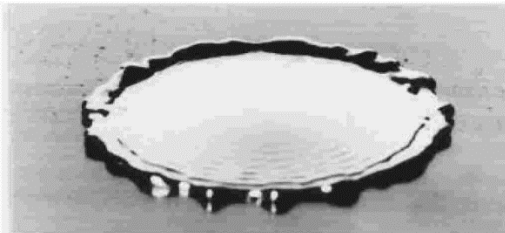
0.1



0.6



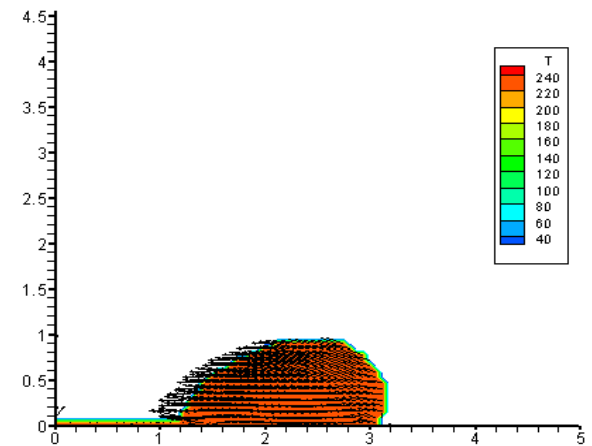
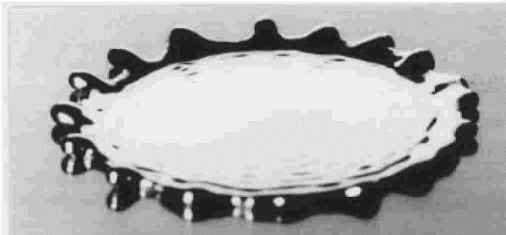
2.0



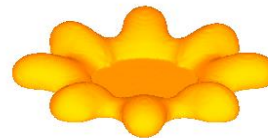
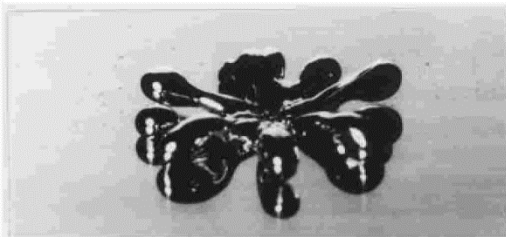
# Surface Tension Work

240°C, 2 m/s

3.2



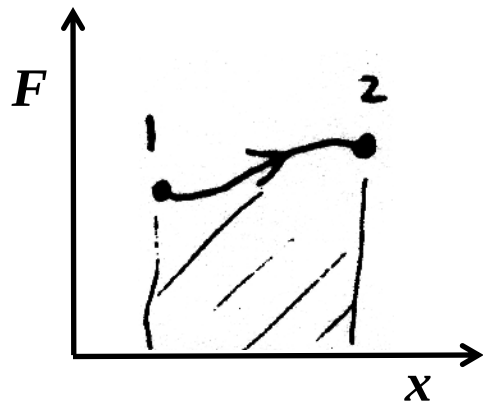
7.0



10.0



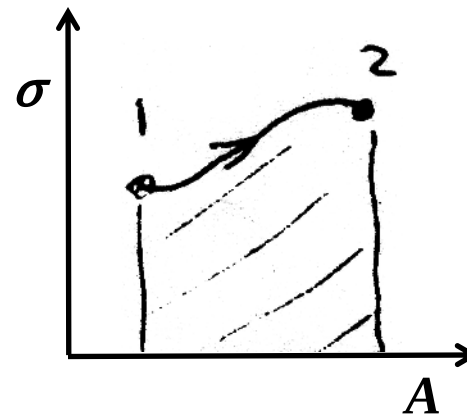
# Various Work Modes



1-D  
energy/distance

$$F \rightarrow N = \frac{N \cdot m}{m}$$

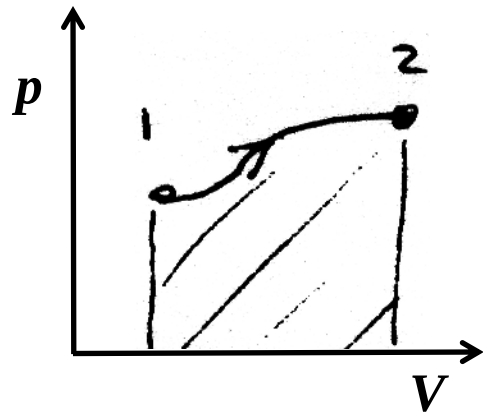
$$W = \int_1^2 F dx$$



2-D  
energy/surface area  
Surface tension

$$\sigma \rightarrow \frac{N}{m} = \frac{N \cdot m}{m^2}$$

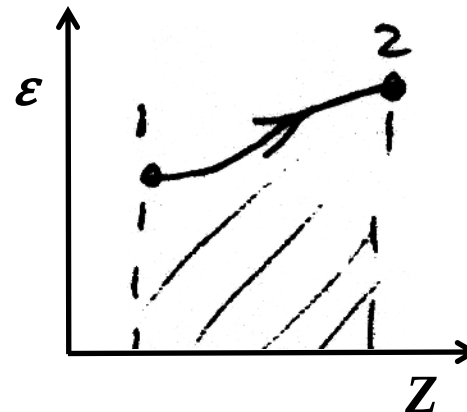
$$W = \int_1^2 \sigma dA$$



3-D  
energy/volume

$$p \rightarrow \frac{N}{m^2} = \frac{N \cdot m}{m^3}$$

$$W = \int_1^2 p dV$$



energy/charge  
Z - charge (coulomb)  
ε - voltage

$$1 \text{ volt} = 1 \frac{\text{joule}}{\text{coulomb}}$$

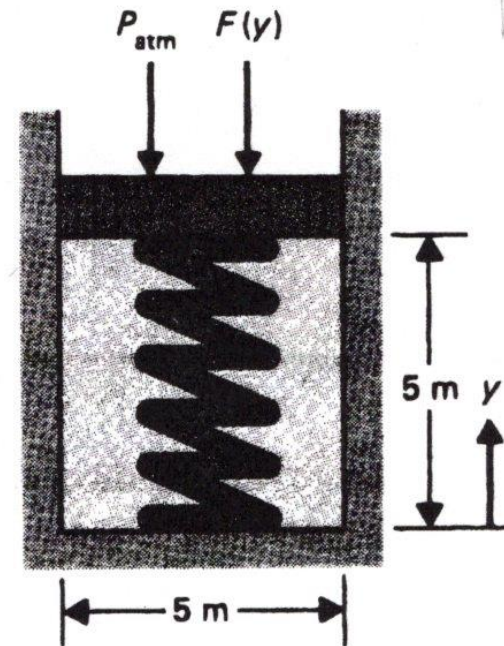
$$W = \int_1^2 \varepsilon dZ$$

Note  $F, \sigma, p, \varepsilon$  - intensive property (define equilibrium)

$x, A, V, Z$  – extensive property (must change to show energy transfer)

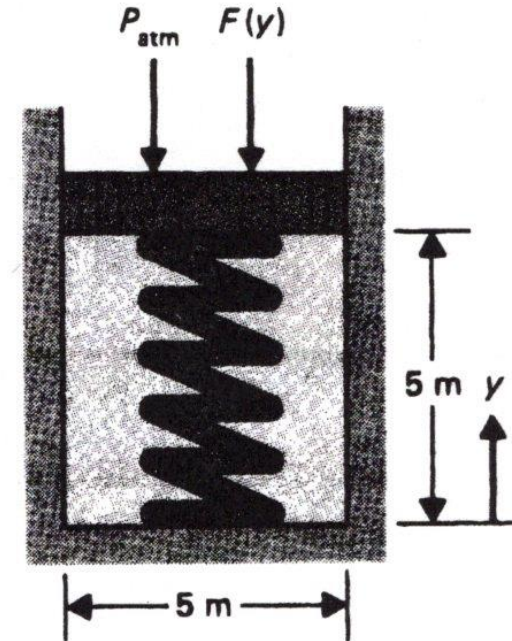
# Work on a Spring Gas System

An ideal gas (air) is contained within a piston-cylinder system as shown. The piston is weightless and frictionless. The atmospheric pressure outside the cylinder is 1 atm (101.3 kPa). Initially, a spring inside the cylinder is held in a compressed state by the piston to a length of 5 m. The spring is then allowed to expand, slowly pushing the piston outward against a restraining force  $F(y)$  until the spring reaches a length of 10 m, where it is fully extended and just barely touches the piston. Throughout this process, the gas in the cylinder is held at constant temperature. The spring constant is  $10^3$  kN/m, and the spring has negligible volume.



- (a) Sketch the P-V diagram for the gas for the process described, showing numerical values for P and V at each end of the process.
- (b) Calculate the work done by the piston on the spring-gas system.

# Example – Shock Absorber



Zimmer Shock Absorbers  
With *Spiral Groove*  
Technology



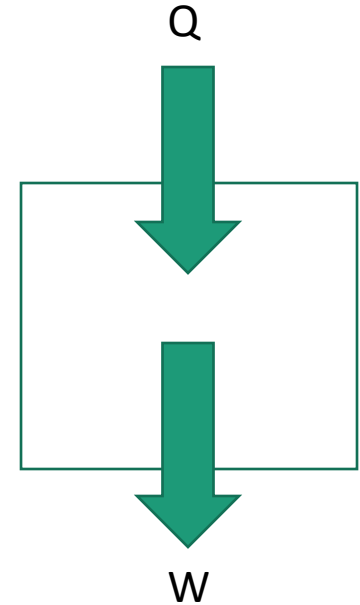
Energy can neither be created or destroyed.

**Energy is CONSERVED!**

# First Law of Thermodynamics for Closed Systems

Process for closed system from State 1 to State 2

$$Q_{1-2} - W_{1-2} = E_2 - E_1$$



Total net **heat added** to system during the process =  $Q_{12}, Q_1^2, {}_1Q_2$

Total net **work done by** the system =  $W_{12}, W_1^2, {}_1W_2$   
(against “forces” in surrounding process)

Total energy in the system at state 1 or 2 =  $E_1$  or  $E_2$

# First Law of Thermodynamics for Closed Systems

Process for closed system from State 1 to State 2

$$Q_{1-2} - W_{1-2} = \Delta KE + \Delta PE + \Delta U$$

Kinetic Energy = KE

$$KE = \frac{1}{2}mv^2$$

Gravitational Potential Energy = PE

Internal Energy in system = U

# Characteristic Properties of a Substance

## Objectives:

- Identify characteristics common to most substances (e.g., gas, liquid, solid, etc.)
- Discuss methods of finding a system's state (i.e., all thermodynamic and related **properties**) by considering property relations for a broad range of substances in tabular, graphical, and equation form.



# Pure, Simple, Compressible Substance

Pure = unchanged chemical composition

Example:

Not pure: Fuel + air ( $N_2 + O_2$ )  $\rightarrow$  combustion gases

Pure: all water (liquid water and steam)

all air (mix of  $N_2$ ,  $O_2$ , ... as long as it does not change)

**Simple = only **one** form of work for energy transfer between system and surroundings**

**Compressible = only work from compression:**

$$W_{12} = \int_{V_1}^{V_2} p(V) dV \quad \text{or} \quad w_{12} = \frac{W_{12}}{m} = \int_{v_1}^{v_2} p(v) dv$$

# Phase

Phase indicates a type of molecular arrangement with the system

- Gas – quite energetic, in constant movement
- Liquid – less energy than gas
- Solid – less energy than liquid, locked in lattice  
denser than liquid, little movement

# State Postulate

For a pure, simple compressible substances only 2 independent, intensive properties are required to define the state (i.e., to find all other properties)  $T, p, v, h, s, u, \text{etc...}$

e.g.  $p = f(v, u)$   $T = f(p, v)$  for ideal gas:  $pv = RT$

During phase changes: **T and P are not independent!**

e.g.  $p, v$  or  $T, v$  define the state of 2-phase

Extensive property can be used to define the state, but **will need 3 properties!**

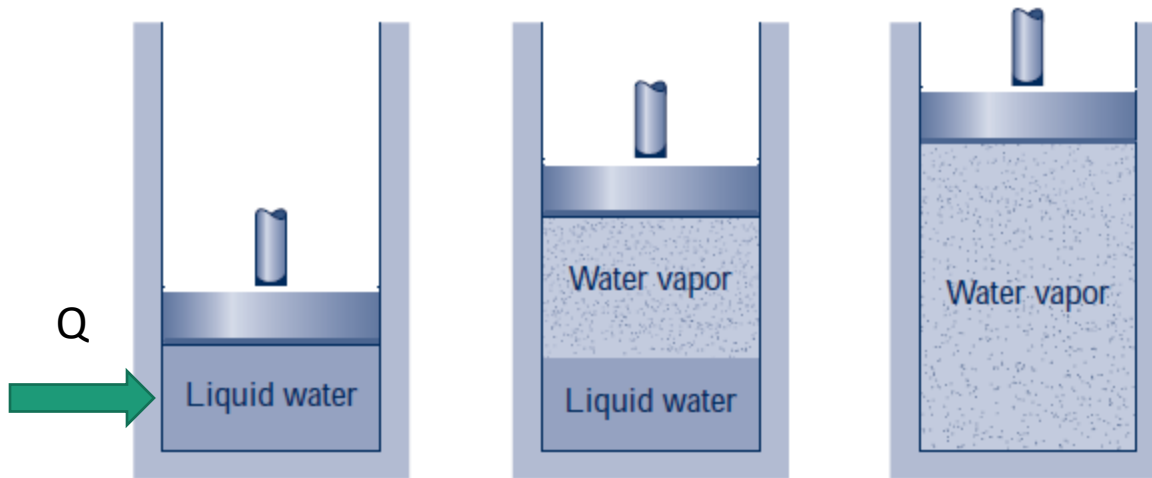
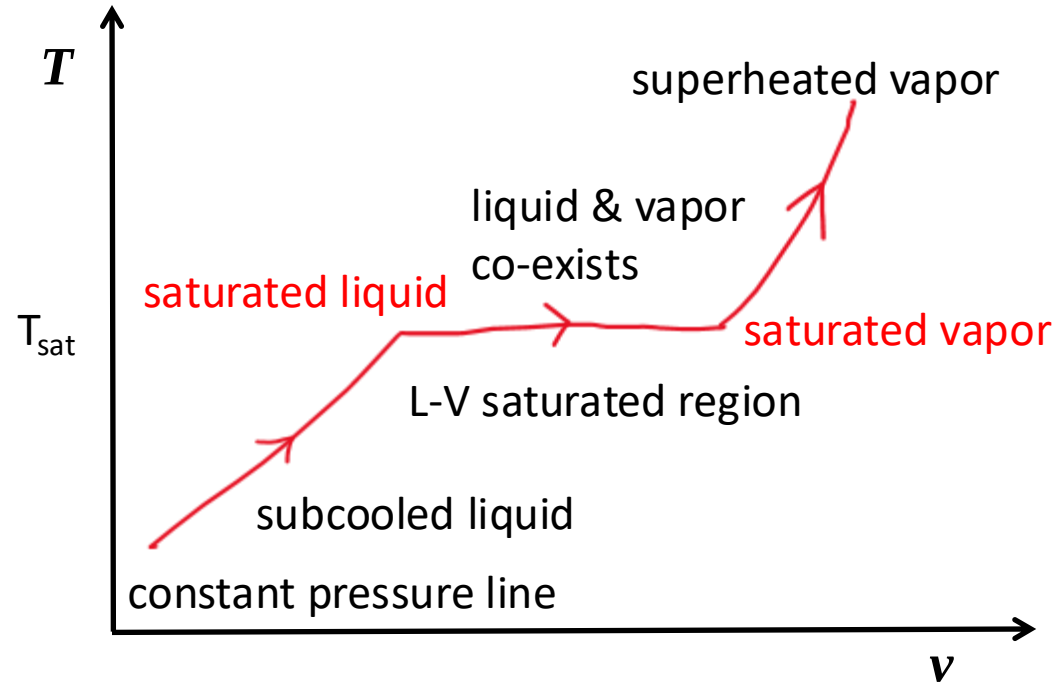
e.g.  $p = f(T, V, m)$   $\xleftrightarrow{\text{equivalent to}}$   $p = f(T, v)$

# $p$ - $v$ - $T$ Relation

Isobaric expansion:

$$p = \frac{\text{weight}}{A} = \text{constant pressure}$$

$$pv = RT$$



# T-v Diagram

No discontinuity during phase change  
Supercritical state

